MILITARY SPECIFICATION

CONTROL, COMMANDERS, LASER RANGING: C-9134( )/VVG-1: 11738811

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of electronic unit known as Control, Commander's, Laser Ranging: C-9134( )/VVG-1: 11738811 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment; Acquisition, Maintenance, and Disposition of
MIL-S-45743	Soldering, Manual Type, High Reliability Electrical, Electronic, Instrument, Communication, and Radar for Aerospace and Control Systems, Procedures for

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MIL-S-46844	Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Envirionmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F8565556	Pressure Tester
F10955915	Assembly, Bearing
F11738811	Control, Commander's, Laser Ranging: C-9134( )/VVG-1

PACKAGING DATA SHEET

11738811	Packaging of Control, Commander's, Laser Ranging: C-9134( )/VVG-1

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Fabrication. The Laser Ranging Commander's Control C-9134( )/VVG-1, herein referred to as the unit, shall be manufactured in accordance with Drawing F11738811 and drawings pertaining thereto.

3.2 General specification. The unit shall meet the following requirements, where applicable, of MIL-F-13926:

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finish

d. Electrical and electronic assemblies

e. Part identification and marking

f. Workmanship.

3.2.1 Soldering. Soldering shall be in accordance with MIL-S-45743 or MIL-S-46844, except that bare copper wire exposure shall be permitted at clipped ends of component leads.

3.3 Performance. Unless otherwise specified, the unit shall meet the performance requirements of this specification at standard ambient temperatures between 60 degrees Fahrenheit (°F) and 90°F.

3.3.1 Loads, power, and signals. The unit shall perform as specified herein when the loads, power, and signals of table I are applied as specified.

3.3.1.1 LAST indicator. With the +5.0 Vdc and +10 Vdc powers (items 2.1 and 2.2) of table I applied between pins J1-K and J1-L, and pins J1-H and J1-L, respectively, the LAST indicator shall not light. Upon subsequent application of the +15 Vdc power (item 2.3) of table I between pins J1-J and J1-L, the LAST indicator shall light.

3.3.1.2 RANGE RETURN SELECTOR. With the RANGE RETURN SELECTOR pushbuttons depressed and released in turn, the voltages specified in table II shall appear at the designated pins of J1.

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TABLE I
LOADS, POWER, AND SIGNALS

Item	Conditions	Characteristics	Connections
1.	Loads		Connected between the following pins:
1.1	Resistor	180 ohms (Ω) ±5 percent	J1-AA and J1-L
1.2	Resistor	180Ω ±5 percent	J1-M and J1-N
1.3	NAND gate	6 gates - - DTμL9946, or equivalent	J1-W and J1-L
2.	Power Sources		Unless otherwise specified, applied between pins:
2.1	+5 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤ 25 mV p-p	J1-K and J1-L
2.2	+10 Vdc	Tolerance: ±0.2 Vdc Ripple: ≤ 25 mV p-p	J1-H and J1-L
2.3	+15 Vdc	Tolerance: ±0.7 Vdc Ripple: ≤ 25 mV p-p	J1-J and J1-L
3.	Signal sources		Applied as specified herein
3.1	Digital Type-A	Logical one: +4.0 ± 1.0 V Logical zero: +0.2 ± 0.2 V	
3.2	Digital Type-B	Logical one: open circuit Logical zero: +0.5 ± 0.5 V	

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TABLE II
RANGE RETURN SELECTION RESPONSES

Item	RANGE RETURN SELECTOR	Outputs At J1(1)		
		t	u	v
1	1	+4.2 ±1.0 Vdc	+0.2 ±0.2 Vdc	+0.2 ±0.2 Vdc
2	2	+0.2 ±0.2 Vdc	+4.2 ±1.0 Vdc	+0.2 ±0.2 Vdc
3	LAST	+0.2 ±0.2 Vdc	+0.2 ±0.2 Vdc	+4.2 ±1.0 Vdc
NOTE 1: Voltages are referenced to J1-L (return).				

3.3.1.3 LASER MODE CONTROL switch. With the LASER MODE CONTROL switch set to each position in turn, the circuit conditions and voltages at the designated terminals shall be as specified in table III.

3.3.1.4 Dimmer (DMR) switch. Depressing the DMR switch shall cause all indicators other than the RANGE (METERS) and RETURNS, to change either from bright to dim or from dim to bright. When the indicators are bright, the voltages at J1-AA and J1-M shall be +13.5 ±1.5 Vdc and +15.0 ±1.5 Vdc, respectively. When the indicators are dim, the voltages at J1-AA and J1-M shall be +8.5 ±2.0 Vdc and +10.0 ±2.0 Vdc, respectively.

3.3.1.5 Test switch (TSW). Depressing the TSW switch shall cause:

a. All indicators to light,

b. Each RANGE (METERS) and RETURNS readout to display the number 8, and

c. The voltage at J1-N to be +0.8 ±0.4 Vdc

3.3.1.6 Indicator illumination

3.3.1.6.1 RANGE RETURN SELECTOR indicators. The 1, 2, and LAST indicators shall light when depressed. Only the indicator depressed last shall be lighted.

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TABLE III
LASER MODE CONTROL SWITCHING RESPONSES

Item	LASER MODE CONTROL Switch Position	Outputs at JI (1), (2)	
		P	W
1	OFF	O	NA
2	TEST	S	+0.2 ±0.2 Vdc
3	ON	S	+4.2 ±1.0 Vdc

Note 1: Outputs are referenced to J1-L (return)
Note 2: Legend - S = Short circuit (≤0.3 ohm)
O = Open circuit (≥ 1 megohm)
NA = Not applicable

3.3.1.6.2 Battery-low/drain (BAT-LOW/DR) and malfunction (MALF) indicators. With the logical zero type-A signal (item 3.1) of table I applied to J1-p, the MALF indicator shall light. With the logical zero type-B signal (item 3.2) of table I applied to J1-S and to J1-CC, the BAT-LOW and BAT-DR indicators, respectively, shall light.

3.3.1.6.3 LASER MODE CONTROL indicators. With the LASER MODE CONTROL switch set to ON or TEST, the ON or TEST indicator, respectively shall light.

3.3.1.7 RETURNS readout selection logic. With the digital type-A signal of table I applied as specified in table IV, the readout number on the RETURNS display shall be as specified in table IV.

3.3.1.8 RANGE (METERS) readout selection logic. With the digital type-A signal (item 3.1) of table I applied as specified in table V, each of the four display positions of the RANGE (METERS) readout shall indicate as specified in table V.

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TABLE IV
RETURNS READOUT SELECTION LOGIC

Inputs to Pins of J1				RETURNS Readout
j	k	m	n	
1	0	0	1	0
0	1	1	0	6

NOTES: 1. All terminals are referenced to J1-L (return).
2. Legend: 1 = logical one type-A
0 = logical zero type-A

3.3.1.9 Range indicator drive. With the logical one digital type-A signal (item 3.1) of table I applied to J1-R, the voltage at J1-N shall be +15.0 ±1.0 Vdc. With the logical zero type-A signal (item 3.1) of table I applied to J1-R, the voltage at J1-N shall be +0.4 ±0.4 Vdc.

3.3.1.10 Readout blanking. With the logical zero digital type-A signal (item 3.1) of table I applied to J1-y, the range readout shall be blanked momentarily.

3.3.1.11 Manual reset signal. With no signal applied to J1-T, the voltages at J1-V and J1-U shall each be +4.2 ±1.0 Vdc. With the logical zero digital type-A signal (item 3.1) of table I applied to J1-T, the voltages at J1-V and J1-U shall each be +0.2 ±0.2 Vdc.

3.3.1.12 Turn-off reset. The time interval between turn-off of the +15 Vdc power (item 2.3) of table I and the change in level of the voltages at J1-V and J1-U shall be as shown in figure 1.

3.3.1.13 Turn-on reset. The time interval between turn on of the +15 Vdc power (item 2.3) of table I and the change in level of the voltages at J1-V and J1-U shall be as shown in figure 2.

3.3.1.14 MALF indicator disable. With the digital type-A signal (item 3.1) of table I applied as specified in table VI, the response of the MALF indicator shall be as specified in table VI.

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TABLE V
RANGE (METERS) READOUT SELECTION LOGIC

Thousands						Hundreds					Tens					Units						
Input to J1(1)(2)					RANGE (METERS)	Input to J1(1)(2)				RANGE (METERS)	Input to J1(1)(2)				RANGE (METERS)	Input to J1(1)(2)						RANGE (METERS)
x	f	g	h	i	Readout	b	c	d	e	Readout	X	Y	Z	a	Readout	x	s	r	q	p	W	Readout
1	0	0	0	1	7	0	0	0	1	7	0	0	0	1	7	1	1	1	1	1	0	5
1	1	1	1	0	8	1	1	1	0	8	1	1	1	0	8	0	1	1	1	0	0	5
0	1	1	1	1	B	-	-	-	-	-	-	-	-	-	-	1	1	1	1	0	1	1
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1	1	0	1	1	2
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1	0	1	1	1	3
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	0	1	1	1	1	4

NOTES: 1. All J1 pins are referenced to J1-L (return).
2. Legend: 1 = logical one, type A
0 = logical zero, type A
B = blank

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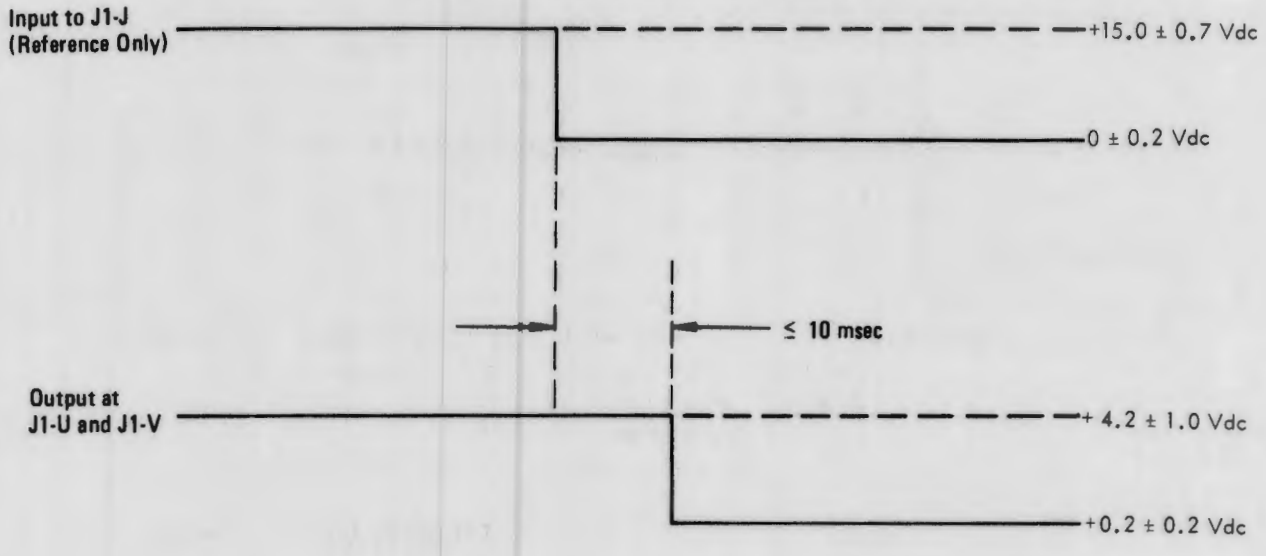
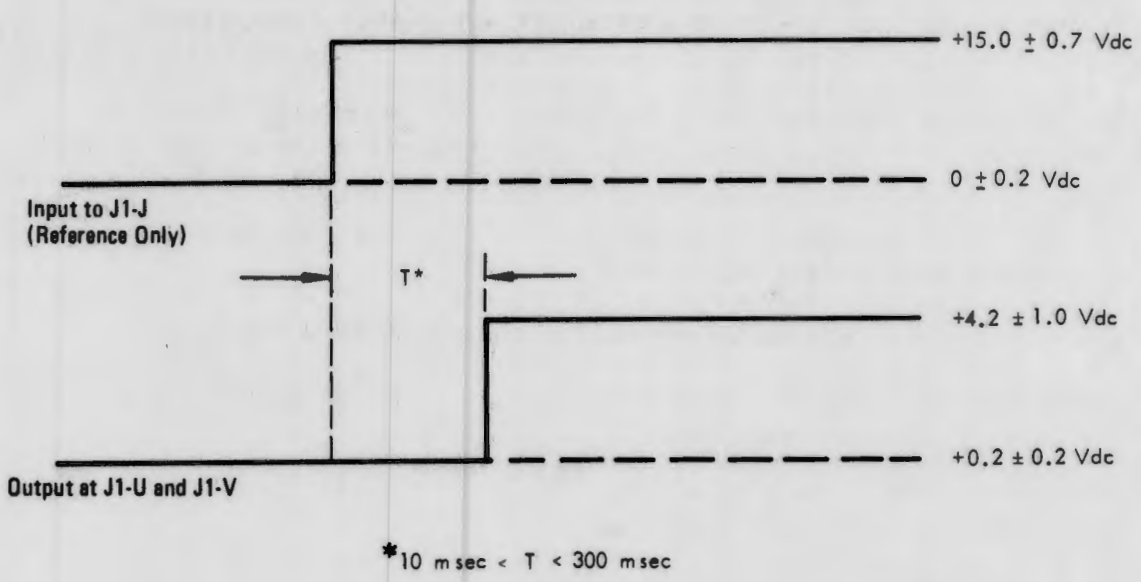


Figure 1. Turn-off Reset Time



*10 msec < T < 300 msec

Figure 2. Turn-on Reset Time

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TABLE VI
MALF INDICATOR DISABLE

Input to J1(1)(2)							MALF Indicator
BB	x	p	q	r	s	W	
NC	1	0	1	1	1	1	Lighted
0	1	0	1	1	1	1	Not lighted

NOTES: 1. All J1 pins are referenced to J1-L (return)
2. Legend: 1 = logical one, type A
2 = logical zero, type A
NC = no connection

3.3.2 Environmental.

3.3.2.1 Storage temperature. The unit shall meet the requirements of 3.3.1 at ambient temperature after exposure to and thermal stabilization at -80° F and +160° F.

3.3.2.2 Operating temperatures. The unit shall meet the requirements of 3.3.1 while exposed to and thermally stabilized at -40°F and +125°F, subsequent to which the requirements of 3.3.1 shall be met at ambient temperature.

3.3.2.3 Vibration. The unit shall show no physical failure and shall meet the requirements of 3.3.1 after being exposed to the following vibratory conditions:

a. Amplitude: 1/64 inch, constant (1/32 inch double amplitude)

b. Vibratory motion: Simple harmonic

c. Sweep time: 5 to 55 to 5 Hertz in a 5- minute period

d. Vibration cycling: Three linear or logarithmic sweeps applied to each of the three mutually perpendicular axes (see figure 3).

3.3.2.4 Shock. The unit shall meet the requirements of 3.3.1 after being exposed to a total of 18 shock impulses, three in each direction along the three mutually perpendicular axes defined on figure 3. Each shock impulse shall be a half-sine wave, with a time duration of 3.5 ±0.5 milliseconds and a peak amplitude of 100g.

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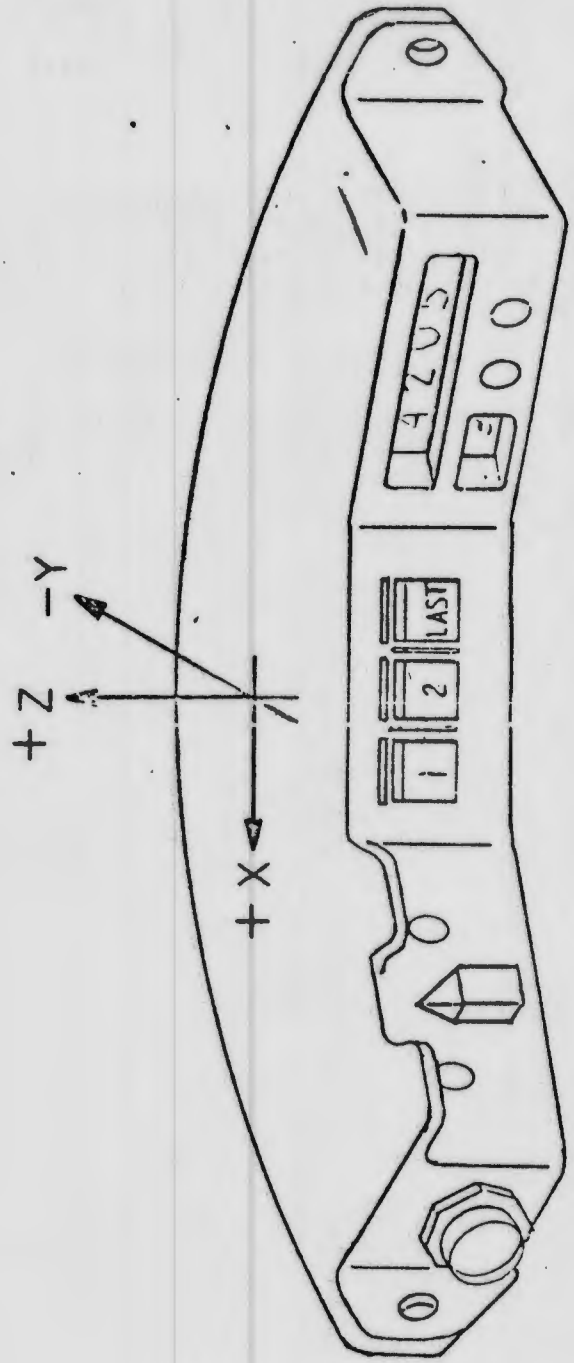


Figure 3. Commanders Control-Axes Definition

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3.3.3 Sealing. With the unit initially pressurized with nitrogen or air to 5 ±1 pounds per square inch gage (psig), the loss of pressure shall not exceed 0.055 psig in one-half hour, except for special sampling-environmental during which the loss of pressure shall not exceed 0.22 psig in 2 hours.

3.3.4 Internal pressure. Subsequent to meeting the sealing requirements, the unit shall be dry flushed with an internal atmosphere containing nitrogen or air having a dew point of at least -63.5°F. After the unit is flushed, an internal pressure of 3.5 ±0.5 psig shall be established.

3.3.5 Interface. The unit shall be capable of interfacing with the Bearing Assembly, F10955915.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three units. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure conformance to all the requirements of this specification. A record of this inspection, including certificates of conformance for materials shall be submitted with the sample.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of this specification. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

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4.3 Inspection provisions

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with inspection and sampling procedures herein and submission of product provisions of MIL-STD-105.

4.3.2 Examination and tests

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in the respective specifications and Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.3.2.2 First article testing. First article testing shall be conducted in accordance with tables VII through IX, to determine compliance with 4.2. All examinations and tests shall be witnessed by the Government Quality Assurance Representative. When manufacturing processes, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability of the unit, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.3.2.3 Function tests. The requirements and tests in table VII shall be inspected on a 100 percent basis. All examinations and tests shall be conducted at the standard ambient temperature.

4.4 Special sampling. The selection of samples shall be in accordance with section 7 of MIL-STD-105.

4.4.1 General. One unit shall be selected at random by a Government representative as a control sample from each 50 produced. The control sample shall meet the requirements and tests in table VIII.

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TABLE VII
FUNCTIONAL TESTS

Classification	Characteristics	Requirements	Test Paragraph
101.	LAST indicator	3.3.1.1	4.6.2
102.	RANGE RETURN SELECTOR	3.3.1.2	4.6.3
103.	LASER MODE CONTROL switch	3.3.1.3	4.6.4
104.	Dimmer (DMR) switch	3.3.1.4	4.6.5
105.	Test (TSW) switch	3.3.1.5	4.6.6
106.	Indicator illumination	3.3.1.6	4.6.7
107.	RETURNS Read-out selection logic	3.3.1.7	4.6.8
108.	RANGE (METERS) readout selection logic	3.3.1.8	4.6.9
109.	Range indicator drive	3.3.1.9	4.6.10
110.	Readout blanking	3.3.1.10	4.6.11
111.	Manual reset signal	3.3.1.11	4.6.12
112.	Turn-off reset	3.3.1.12	4.6.13
113.	Turn-on reset	3.3.1.13	4.6.14
114.	MALF indicator disable	3.3.1.14	4.6.15
115.	Sealing (1/2 hour)	3.3.3	4.6.17

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TABLE VIII
SPECIAL SAMPLING - GENERAL

Classification	Control Tests	Requirements	Test Paragraph
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental. Three units shall be selected at random by a Government representative as a control sample from each 50 produced or from each month's production, whichever occurs first. The control samples shall meet the requirements and tests in table IX.

4.4.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with the acceptance and rejection criteria specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607. The inspection equipment listed in table X shall be utilized in performing the tests specified herein.

TABLE IX
SPECIAL SAMPLING - ENVIRONMENTAL

Classification	Control Tests	Requirements	Test Paragraph
303.	Storage temperature	3.3.2.1	4.6.16.1
304.	Operating temperature	3.3.2.2	4.6.16.1
305.	Vibration	3.3.2.3	4.6.16.2
306.	Shock	3.3.2.4	4.6.16.3
307.	Sealing (2 hours)	3.3.3	4.6.17
308.	Internal pressure	3.3.4	4.6.18
309.	Interface	3.3.5	4.6.19

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TABLE X
INSPECTION EQUIPMENT

Item	Nomenclature	Description
1.	DC Power Supply	+5.0 ±0.2 Vdc Ripple: 25 mV p-p maximum
2.	DC Power Supply	+10.0 ±0.2 Vdc Ripple: 25 mV p-p maximum
3.	DC Power Supply	+15.0 ±0.7 Vdc Ripple: 25 mV p-p maximum
4.	Ohmmeter	Range: 0 to 10 megohms Accuracy: ±5 percent
5.	DC Voltmeter	Range: 0 to 20 Vdc Accuracy: ±1 percent
6.	Signal Source	Capable of providing d-c levels of amplitude as specified in items 3.1 and 3.2, table I.

4.5.1 Government-furnished inspection equipment. Where the contract provides for Government furnished inspection equipment, supply and maintenance of inspection equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Contractor-furnished inspection equipment

4.5.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAPs forming a part of the contract shall be supplied by the contractor in accordance with applicable technical data listed in the TDP.

4.5.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the test methods and procedures specified in 4.6 of this specification and with the component inspection procedures specified in examination and test facilities provisions of MIL-F-13926. Since tolerance of inspection equipment is normally considered to be within 10 percent of the product tolerance for which intended, this inherent error in the inspection equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of inspection equipment shall be so selected and controlled

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as to insure that the inspection equipment shall reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of inspection equipment.

4.5.2.2.1 Pressure tester. Special test equipment (STE) in accordance with Drawing F8565556, or equivalent, shall be provided to check the sealing of the unit.

4.6 Test methods and procedures

4.6.1 Test conditions. Unless otherwise specified, the test conditions shall be in accordance with the test facilities provisions of MIL-F-13926.

4.6.1.1 Initial conditions. Connect the loads and power to the unit as specified in table I.

4.6.2 LAST indicator test. Prior to and after application of power, item 2.3 of table I, observe the lighting of the LAST indicator to determine compliance with 3.3.1.1.

4.6.3 RANGE RETURN SELECTOR test. Depress and release each RANGE RETURN SELECTOR pushbutton listed in table II. Measure the output voltages at the associated pins specified in table II to determine compliance with 3.3.1.2.

4.6.4 LASER MODE CONTROL switch test. With the power off, turn the LASER MODE CONTROL switch to each position (table III). Measure the resistance between J1-P and J1-L for each switch position. With the power on, measure the voltage at J1-w with respect to J1-L for each applicable switch position. Verify compliance with 3.3.1.3.

4.6.5 DMR switch test. Depress the DMR switch and observe all indicators, except the RANGE (METERS) and RETURNS indicators. Measure the voltage at J1-AA and J1-M. Verify that the requirements of 3.3.1.4 are met. Again depress the DMR switch and observe the indicators. Again measure the voltage at J1-AA and J1-M. Verify that the requirements of 3.3.1.4 are met.

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4.6.6 TSW switch test. Depress the TSW switch. Observe all indicators and the RANGE (METERS) and RETURNS numeric displays. Measure the voltage at J1-N, to determine compliance with 3.3.1.5.

4.6.7 Indicator illumination

4.6.7.1 RANGE RETURN SELECTOR indicators test. Depress each RANGE RETURN SELECTOR pushbutton (1, 2 and LAST). Observe the associated indicator to determine compliance with 3.3.1.6.1.

4.6.7.2 BAT-LOW/DR and MALF indicators test. Apply the logical zero digital type-A signal of table I to J1-p and the logical zero digital type-B signal of table I to J1-S and J1-CC. Observe operation of the MALF, BAT-LOW, and BAT-DR indicators to determine compliance with 3.3.1.6.2.

4.6.7.3 LASER MODE CONTROL indicators test. Turn the LASER MODE CONTROL switch to each of positions ON and TEST. Observe operation of the ON and TEST indicators to determine compliance with 3.3.1.6.3.

4.6.8 RETURNS readout selection logic test. Apply the type-A signal of table I to the group of J1 terminals as specified in table IV. Observe the RETURNS readout. Verify that the requirement of 3.3.1.7 is met.

4.6.9 RANGE (METERS) readout selection logic test. Apply the type-A signal of table I to the group of J1 terminals, in turn, under the thousands, hundreds, tens and units columns of table V. Observe the RANGE (METERS) readout. Verify that the requirement of 3.3.1.8 is met.

4.6.10 Range indicator drive test. Apply the type-A signal of table I to J1-R at the logical one level, then at the logical zero level. Measure the voltage at J1-N for each input signal condition to determine compliance with 3.3.1.9.

4.6.11 Readout blanking test. Apply the logical zero digital type-A signal of table I to J1-y. Observe the range readout to determine compliance with 3.3.1.10.

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4.6.12 Manual reset signal test. With no signal applied to J1-T, measure the voltages at J1-V and J1-U. Apply the logical zero digital type-A signal of table I to J1-T, and remeasure these voltages. The voltages at J1-V and J1-U shall comply with 3.3.1.11.

4.6.13 Turn-off reset test. Turn off the +15 Vdc power to J1-J. Measure the time interval between the +15 Vdc power turn-off and the change in the voltage levels at J1-V and J1-U. Verify compliance with 3.3.1.12.

4.6.14 Turn-on reset test. Turn off and then turn on the +15 Vdc power at J1-J. Measure the time interval between the +15 Vdc power turn-on and the change in voltage levels at J1-V and J1-U. Verify compliance with 3.3.1.13.

4.6.15 MALF indicator disable test. Apply the type-A signals of table I to the connector pins listed in table VI. Observe the MALF indicator. Verify that the requirements of 3.3.1.14 are met.

4.6.16 Environmental tests

4.6.16.1 Temperature test. The test apparatus shall be in accordance with the applicable test facilities provisions of MIL-F-13926 except that the unit may be mounted on the inside surface of the test chamber. Perform this test in accordance with Procedure I of MIL-F-13926, with conditions C (-40°F) and E (+125°F), and with the rate of temperature change during cycling not greater than 8°F per minute. At each of the specified operating temperatures, perform the tests of 4.6.2 through 4.6.15 to determine compliance with 3.3.2.1 and 3.3.2.2.

4.6.16.2 Vibration tests. Perform this test utilizing the apparatus specified in Method 514.1 of MIL-STD-810 and in accordance with Part 1 of Procedure I therein, except that the vibratory conditions of 3.3.2.3 shall apply. Following vibration exposure, examine the unit for structural failures, loose hardware, wire breakage, and abrasion. Perform the tests of 4.6.2 through 4.6.15 to determine compliance with 3.3.2.3.

4.6.16.3 Shock test. Perform this test utilizing apparatus specified in Method 516.1 of MIL-STD-810, in accordance with Procedure I therein and with the conditions specified in 3.3.2.4, to determine compliance with 3.3.2.4.

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4.6.17 Sealing test. Perform this test utilizing STE specified in 4.5.2.2.1. Pressurize the unit as specified in 3.3.3. Measure the pressure loss over the applicable time period to determine compliance with 3.3.3.

4.6.18 Internal pressure. Perform this test utilizing the STE specified in 4.5.2.2.1. Verify that the requirements of 3.3.4 are met.

4.6.19 Interface test. This test shall be conducted using a limit gage to determine compliance with 3.3.5.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11738811. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Laser Ranging Commanders Control C-9134( )/VVG-1 is a functional unit of the Fire Control Rangefinder AN/VVG-1( ). The unit provides the following:

a. Operational conditions indications,

b. Selection of range return,

c. Range and returns displays, and

d. Control logic functions.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Applicable packaging data sheet number (see 5.1)

c. Selection of applicable levels of preservation, packaging, and packing

d. Applicable stock number

e. Requirement for First Article Submission.

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Preparing Activity:

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